

Urban AI: AI-Driven Decision Support for Real-World Urban Challenges

Fall 2026 · MSU AI-Ready Initiative · Non-credit Learning Experience

Time: Monday & Wednesday 5:30 pm – 7:30 pm, October 28 – November 23, 2026 (16 hours)

Location: <https://msu.zoom.us/j/94058399715>

Format: Free · Non-credit · Online · Open to all majors · No coding required

Registration: Seats are limited. Register by Friday, October 23, 2026 at <https://jiehou-lab.github.io/urban-ai>

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Course Website: <https://jiehou-lab.github.io/urban-ai/>

COURSE OVERVIEW

Urban AI is a 16-hour online, non-credit learning experience, open to students from all majors, that introduces AI-driven decision support systems (DSSs) in the context of real-world urban development and planning challenges. Decisions about cities increasingly involve large volumes of spatial, environmental, and socioeconomic information. Planners, public administrators, real estate developers, and community representatives are expected to decide under uncertainty, communicate tradeoffs, and use digital tools responsibly to build consensus. AI is already part of that work: computer vision analyzes satellite and street-level imagery to assess land use and building conditions, text mining turns public comments and geolocated social media into a picture of community concerns, and large language models (LLMs) help generate scenario analyses and draft communications for stakeholders. The same tools also bring risks of bias, privacy loss, limited transparency, uneven data quality, and the exclusion of affected communities.

This course helps students build practical AI literacy, decision-support judgment, and ethical reasoning to work inside that process. Students explore the concept of Urban AI, how planning decisions actually get made, the history and evolution of DSSs from traditional systems to AI-enhanced workflows, and how to evaluate AI outputs before anyone acts on them. The course is organized into four sections: (1) urban analytics and AI concepts; (2) analytic models and cases, including site analysis, simulation, prediction, and scenario testing; (3) decision support systems, examined through deployed cases and the design of AI-enhanced workflows with human checkpoints; and (4) language models and AI ethics. Throughout, the emphasis is on responsible AI use, human supervision, transparency, equity, and stakeholder-centered decision-making, with AI treated as a resource that supports human judgment rather than replaces it.

There are no prerequisites. The course assumes no background in planning, AI, GIS, coding, or data science.

Learning objectives:

By the end of this course, students will be able to:

- 1) Define Urban AI and AI-driven decision support systems, and explain how they can support evidence-based urban development decisions.
- 2) Describe typical decision-making processes in planning, including problem framing, stakeholder identification, decision criteria, alternative development, impact analysis, and human judgment.

- 3) Use approved AI tools responsibly to support urban analytic tasks such as processing information, comparing alternatives, and engaging stakeholders.
- 4) Evaluate AI-generated outputs for accuracy, bias, uncertainty, missing context, privacy risks, and relevance to a specific urban issue.

REQUIRED TEXT

There is no required textbook and no required reading. Course slides and lab instructions will be posted to the course Teams site.

CLASS FORMAT AND REQUIREMENTS

The course meets eight times, twice a week for four weeks, for two hours per meeting. Each meeting pairs a concept class with a hands-on lab/discussion forum.

Two discussion forums anchor the course. Discussion Forum A (launching in Meeting 1) asks where Urban AI is already making decisions in the real world. Discussion Forum B (Meeting 8) asks how human-AI interactions will change decision-making processes in planning. Students also keep an AI Use Log for the full four weeks and complete an AI readiness pre-survey and post-survey.

COMPLETION AND FEEDBACK

Expectations

This is a non-credit learning experience, so there are no letter grades. Students are expected to finish the following tasks:

Component	Expectation
Attendance	Attend the meetings
AI Use Log	Keep the log for all four weeks and submit it with the brief
Discussion Forums A & B	Post and contribute to the discussions
Pre- and post-survey	Complete the AI readiness surveys in Meetings 1 and 8
Lab write-ups and outputs	Encouraged but not required; the teaching team gives feedback on anything submitted

Participation

Learning is a social process and a collective endeavor, so your primary responsibility is active participation: in the concept-class check-ins, and the lab breakouts. Submitting short write-ups and lab outputs is the fastest way to get feedback from the teaching team.

COURSE POLICIES

Technical Assistance

All labs run in a web browser; nothing needs to be installed. If you need technical assistance at any time during the course or to report a problem you can:

- Visit the [MSU Technology support site](#)
- Visit the [Distance Learning Services Support Site](#)
- Or call Distance Learning Services: (800) 500-1554 or (517) 355-2345

Resource Center for Persons with Disabilities (RCPD)

- To make an appointment with a specialist, contact: (517) 353-9642 or TTY: (517) 355-1293

- Web site for RCPD: <http://MYProfile.rcpd.msu.edu>

Commit to Integrity: Academic Honesty

Article 2.3.3 of the [Academic Freedom Report](#) states that "The student shares with the faculty the responsibility for maintaining the integrity of scholarship, grades, and professional standards." In addition, the SPDC and CMSE adhere to the policies on academic honesty as specified in General Student Regulations 1.0, Protection of Scholarship and Grades; the all-University Policy on Integrity of Scholarship and Grades; and Ordinance 17.00, Examinations. (See [Spartan Life: Student Handbook and Resource Guide](#) and/or the MSU Web site: www.msu.edu.)

Although this experience is non-credit, the same expectations apply. You are expected to develop original work with your team, to credit the sources and data you use, and to use AI tools only as described in the AI Use Policy above, with every use documented in your AI Use Log. Contact the instructors if you are unsure about the appropriateness of your work.

The Spartan Code of Honor academic pledge embodies the principles of integrity that every Spartan is required to uphold in their time as a student, and beyond. It was formally adopted by ASMSU on March 3, 2016, endorsed by Academic Governance on March 22, 2016, and recognized by the Provost, President, and Board of Trustees on April 15, 2016.

"As a Spartan, I will strive to uphold values of the highest ethical standard. I will practice honesty in my work, foster honesty in my peers, and take pride in knowing that honor is worth more than grades. I will carry these values beyond my time as a student at Michigan State University, continuing the endeavor to build personal integrity in all that I do." <https://honorcode.msu.edu/>

Limits to Confidentiality

Essays, journals, and other materials submitted for this course are generally considered confidential pursuant to the University's student record policies. However, students should be aware that University employees, including instructors, may not be able to maintain confidentiality when it conflicts with their responsibility to report certain issues to protect the health and safety of MSU community members and others. As instructors, we must report the following information to other University offices (including the Department of Police and Public Safety) if you share it with us:

- Suspected child abuse/neglect, even if this maltreatment happened when you were a child,
- Allegations of sexual assault or sexual harassment when they involve MSU students, faculty, or staff, and
- Credible threats of harm to oneself or to others.

These reports may trigger contact from a campus official who will want to talk with you about the incident that you have shared. In almost all cases, it will be your decision whether you wish to speak with that individual. If you would like to talk about these events in a more confidential setting you are encouraged to make an appointment with the MSU Counseling Center.

Inform Your Instructor of Any Accommodations Needed

From the Resource Center for Persons with Disabilities (RCPD): Michigan State University is committed to providing equal opportunity for participation in all programs, services and activities. Requests for accommodations by persons with disabilities may be made by contacting the Resource Center for Persons with Disabilities at 517-884-RCPD or on the web at rcpd.msu.edu. Once your eligibility for an accommodation has been determined, you will be issued a Verified Individual Services Accommodation ("VISA") form. Because this course runs for only four weeks, please share this form with the instructors before the cohort begins on October 28 so that support can be arranged in advance. Requests received later will be honored whenever possible.

Commercialized Lecture Notes

Commercialization of lecture notes, recordings, and university-provided course materials is not permitted in this course.

Internet

Course materials and selected team briefs may be shared on the course website for educational use. Please notify the instructors in writing if you do not want your work or your team's brief posted.

Disruptive Behavior

Article 2.III.B.4 of the [Academic Freedom Report \(AFR\)](#) for students at Michigan State University states: "The student's behavior in the classroom shall be conducive to the teaching and learning process for all concerned." Article 2.III.B.10 of the AFR states that "The student has a right to scholarly relationships with faculty based on mutual trust and civility." [General Student Regulation 5.02](#) states: "No student shall . . . interfere with the functions and services of the University (for example, but not limited to, classes . . .) such that the function or service is obstructed or disrupted." These expectations apply equally to live online sessions, breakout rooms, chat, and discussion forums. Students whose conduct adversely affects the learning environment may be subject to disciplinary action through the Student Judicial Affairs office.

Attendance

Only registered participants may attend the live sessions. Attendance at a minimum of 7 of the 8 meetings is required to receive the certificate of completion. Recordings are provided so that a missed meeting can be caught up on, but they do not substitute for attendance.

Build Rapport

If you find that you have any trouble keeping up with the labs, your team, or other aspects of the course, let the instructors know as early as possible. With only four weeks, small problems grow quickly. Building rapport and effective relationships is key to becoming an effective professional, so be proactive in telling us when difficulties arise so that we can help you find a solution.

Submitting Work

Discussion posts, lab outputs, the AI Use Log, and the team brief will be submitted through the course Teams site unless otherwise instructed.

Collection of Student Work: For the purposes of program evaluation of the MSU AI-Ready Initiative, survey responses and some student work may be retained for future use without any personal identification or information on the material.

COURSE TOPICS & READINGS

The course is divided into four sections: 1) Urban Analytics and AI Concepts, 2) Models and Cases, 3) Decision Support Systems, and 4) Language Models and Ethics.

We will make every effort to stick to the course schedule, but variations are inevitable.

Week	Date	Topic of class
1	28-Oct	Welcome & What Is Urban AI? Orientation (30 min): introduction of learning experience, privacy and data rules, AI Use Log Class 1-1 (60 min): Urban analytics and smart cities to urban AI; how planning decisions get made; what a decision support system is Discussion Forum A launches (30 min): where is Urban AI already deciding things?
	02-Nov	AI Basics for Urban Problems Class 1-2 (60 min): data, features, models, training and inference; machine learning Lab (60 min): clustering analysis

Week	Date	Topic of class
2	04-Nov	Urban Analytic Models I Class 2-1 (60 min): urban data sources, site analysis Lab (60 min): spatial analysis and clustering
	09-Nov	Urban Analytic Models II Class 2-2 (60 min): AI-augmented urban analytic models Lab (60 min): time series analysis
3	11-Nov	Decision Support Systems I Class 3-1 (60 min): decision support systems (DSS) data, model, design, and application Lab (60 min): risk-response DSS
	16-Nov	Decision Support Systems II Class 3-2 (60 min): AI-enhanced decision support systems Lab (60 min): DSS AI Lab
4	18-Nov	Language Models in Decision Support Class 4-1 (60 min): what language models are useful in decision-making process Lab (60 min): text mining and clustering on planning documents
	23-Nov	AI Ethics, and Wrap-Up Class 4-2 (45 min): AI ethics: bias, equity, privacy, transparency, accountability Discussion Forum B & reflection (30 min): how human-AI interactions will change decision-making processes in planning; conclusion of the learning experience